

Problem E. Steak Sides

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

A steakhouse serves n kinds of steak, numbered 1 to n , and k kinds of side dishes, numbered 1 to k . It offers m set meals. Set meal i contains every steak from l_i to r_i (inclusive), together with one side dish d_i . You want to order some set meals so that every steak $1, 2, \dots, n$ is included in at least one of them. However, you do not want the same side dish twice: **no two ordered set meals may have the same side dish**.

Find the minimum number of set meals to order, or report that it is impossible.

Input

The first line contains three integers n , k and m ($1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq k \leq 20$). The i -th of the next m lines contains three integers l_i , r_i and d_i ($1 \leq l_i \leq r_i \leq n$, $1 \leq d_i \leq k$).

Output

Print the minimum number of set meals, or -1 if it is impossible.

Examples

standard input	standard output
5 3 4 1 4 1 5 5 1 1 2 2 3 5 3	2
3 1 2 1 1 1 2 3 1	-1

Note

In the first example, set meals 1 and 2 would cover all steaks, but both come with side dish 1. Ordering set meals 3 and 4 (or 1 and 4) works.

In the second example, both set meals come with side dish 1, so at most one can be ordered, and neither covers all steaks.